

Goji (*Lycium barbarum* and *L. chinense*): Phytochemistry, pharmacology and safety in the perspective of traditional uses and recent popularity.



Since the beginning of this century, Goji berries and juice are being sold as health food products in western countries and praised in advertisements and in the media for well-being and as an anti-aging remedy. The popularity of Goji products has rapidly grown over the last years thanks to efficient marketing strategies. Goji is a relatively new name given to *Lycium barbarum* and *L. chinense*, two close species with a long tradition of use as medicinal and food plants in East Asia, in particular in China. While only *L. barbarum* is officinal, the fruit (fructus Lycii) and the root bark (cortex Lycii radices) of both species are used in the folk medicine. We review here the constituents, pharmacology, safety, and uses of *L. barbarum* and *L. chinense* with consideration to the different parts of the plant. Investigations of the fruit have focused on proteoglycans, known as " *Lycium barbarum* polysaccharides", which showed antioxidative properties and some interesting pharmacological activities in the context of age related diseases such as atherosclerosis and diabetes. As to the root bark, several compounds have demonstrated a hepatoprotective action as well as inhibitory effects on the rennin/angiotensin system which may support the traditional use for the treatment of hypertension. While there are no signs of toxicity of this plant, two cases of possible interaction with warfarin point to a potential risk of drug interaction. In view of the available pharmacological data and the long tradition of use in the traditional Chinese medicine, *L. barbarum* and *L. chinense* certainly deserve further investigation. However, clinical evidences and rigorous procedures for quality control are indispensable before any recommendation of use can be made for Goji products. Copyright Georg Thieme Verlag KG Stuttgart . New York.